

Healthcare Insurance Claim Review Assistant

Problem Statement

Healthcare insurance claims are traditionally reviewed manually by claim adjudicators. This process is time-consuming, error-prone, and difficult to scale when large volumes of claims are received daily.

Major challenges include:

- Manual verification of claim documents
- Fraudulent claim detection
- Policy coverage validation
- Duplicate claim identification
- High operational costs
- Slow claim processing and approvals

The objective of this project is to build an intelligent AI-driven workflow that automates healthcare insurance claim analysis, validation, fraud detection, policy verification, risk assessment, decision making, and explainability.

Project Objective

The objective of this system is to automate the end-to-end healthcare insurance claim review process by extracting structured information from claim documents, validating medical and financial consistency, detecting fraud patterns, retrieving policy coverage details using RAG, identifying duplicate claims, assessing overall claim risk, generating approval or rejection decisions, providing human-readable explanations, and supporting human approval workflows for critical claims.

High-Level Architecture

The system follows a multi-agent AI workflow architecture.

Workflow:

START

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OCR

↓

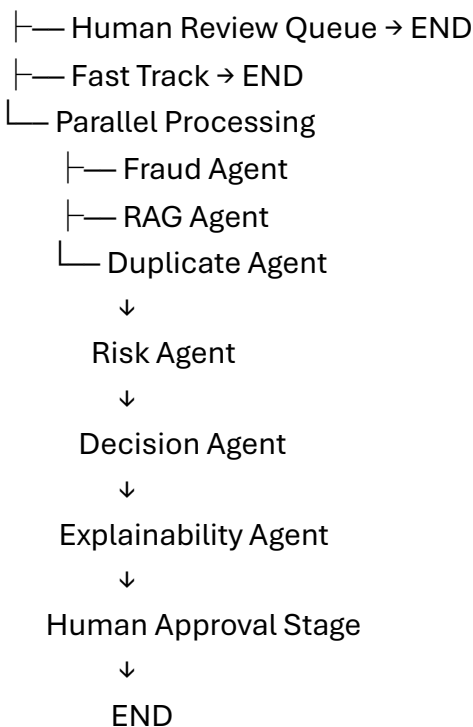
Structuring

↓

Validation

↓

Router



6. Detailed Workflow Explanation

6.1 OCR Agent

The workflow starts with the OCR Agent.

This agent extracts raw text from uploaded healthcare claim PDFs using PyMuPDF. The extracted text contains patient details, diagnosis information, hospital details, treatment data, billing information, provider information, and policy details.

The OCR stage converts unstructured PDF documents into machine-readable text.

Structuring Agent

The Structuring Agent converts raw OCR text into structured JSON format.

The system extracts and organizes all claim-related fields

structured data becomes the central input for all downstream agents.

Validation Agent

The Validation Agent performs both rule-based and semantic validation.

Rule-Based Validation

The system checks:

- Missing mandatory fields
- Invalid dates
- Invalid numeric values
- Claim amount vs policy limits
- Required ICD-10/CPT codes
- Invalid age calculations

Semantic Validation using LLM

The LLM performs intelligent healthcare consistency checks such as:

- Diagnosis matching ICD-10 codes
- Treatment matching CPT codes
- Provider specialty alignment
- Age/gender appropriateness for diagnosis
- Financial consistency checks
- Clinical appropriateness analysis

Fallback logic is implemented in case LLM validation fails.

Router Agent

The Router Agent determines the next processing path based on:

- Validation confidence
- Error severity
- Claim amount
- Fraud indicators
- Missing information
- Risk factors

The Router supports three routes:

Human Review Queue

Critical or suspicious claims are escalated directly for manual review.

Fast Track

Low-risk claims with high confidence are automatically approved for faster processing.

Parallel Processing Pipeline

Medium-risk claims continue through deeper AI analysis.

Fallback routing logic is also implemented.

Parallel Processing Pipeline

To improve performance and reduce processing time, multiple agents run concurrently.

Fraud Agent

The Fraud Agent analyzes:

- Financial fraud patterns
- Suspicious claim amounts
- Near-policy-limit claims
- Invalid provider behavior
- Diagnosis-treatment mismatches
- Backdated claims
- Duplicate patient activity
- Provider specialty inconsistencies

The agent generates:

- Fraud score
- Fraud indicators
- Fraud risk level
- Fraud reasoning

Both LLM-based and rule-based fallback fraud detection are implemented.

RAG Agent (Policy Retrieval)

The RAG Agent retrieves policy clauses relevant to the claim.

Why RAG?

Insurance policy documents are large and complex. Instead of hardcoding all rules, the system dynamically retrieves relevant clauses using semantic search.

Policy Ingestion

Policy PDFs are:

- Extracted using PyMuPDF
- Split into chunks
- Embedded using HuggingFace embeddings
- Stored in ChromaDB

Why ChromaDB?

ChromaDB was selected because:

- It supports persistent vector storage
- Simple integration with LangChain
- Built-in metadata handling
- Easier document management compared to FAISS

RAG Analysis

The agent checks:

- Coverage eligibility
- Policy exclusions
- Claim type coverage
- ICD/CPT coverage
- Coverage percentage
- Estimated payout
- Patient responsibility

Fallback keyword-based retrieval is also implemented.

Duplicate Agent

The Duplicate Agent identifies duplicate claims using similarity scoring.

The system compares:

- Patient details
- Policy numbers
- Diagnosis
- Treatment
- Dates
- Claim amounts
- ICD/CPT codes

Weighted similarity scoring is used to detect duplicate submissions.

. Risk Agent

The Risk Agent combines outputs from:

- Validation Agent
- Fraud Agent
- RAG Agent
- Duplicate Agent

It calculates:

- Overall risk score
- Severity level
- Risk factors
- Processing confidence

This creates a consolidated risk profile for the claim.

Fallback risk assessment logic is also available.

Decision Agent

The Decision Agent acts as the final AI adjudicator.

It analyzes:

- Validation results
- Fraud score
- Policy coverage
- Risk score
- Duplicate detection
- Financial compliance

Possible outcomes:

- APPROVED
- REJECTED
- ESCALATE

The agent also generates:

- Decision confidence
- Reasoning
- Recommended action

Fallback decision rules are implemented for reliability.

Explainability Agent

The Explainability Agent converts technical AI decisions into human-readable explanations.

The agent generates:

- Decision summaries
- Key influencing factors
- Policy references
- Coverage explanations
- Next-step recommendations

This improves transparency and trust in AI decisions.

Fallback explanation generation is also supported.

Human Approval Stage

After the AI decision and explainability stages, a Human Approval stage was added in the UI.

This allows claim reviewers or adjudicators to:

- Review AI-generated decisions
- Verify fraud/risk findings
- Check policy coverage analysis
- Override decisions if necessary
- Approve or reject claims manually

This ensures human oversight for sensitive healthcare claims.

Audit Trail

Throughout the workflow, every agent records audit entries including:

- Processing timestamps
- Agent outputs
- Decision reasoning
- Validation results
- Fraud analysis
- Risk assessments

This provides:

- Traceability
 - Compliance
 - Monitoring
 - Debugging support
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Fallback Mechanisms

Fallback logic is implemented in every major stage of the workflow.

If any LLM call fails, the system automatically switches to:

- Rule-based validation
- Keyword-based policy retrieval
- Rule-based fraud detection
- Deterministic risk scoring
- Rule-based decision making

This ensures high system reliability and uninterrupted processing.